

Experiment 14: Exponentiation

Aim

To implement the **Exponentiation** algorithm using the **Divide and Conquer** technique in Python to calculate a^n .

Algorithm

Step 1: Start.

Step 2: Read the base a and exponent n.

Step 3: If $n = 0$, return 1.

Step 4: Recursively compute the power of a for $n // 2$.

Step 5:

- If n is even, return $\text{half} \times \text{half}$.
- If n is odd, return $a \times \text{half} \times \text{half}$.

Step 6: Display the result.

Step 7: Stop.

Source Code (Python)

```
main.py  [Icons] [Share] [Run]

1  # Exponentiation using Divide and Conquer
2
3  def power(a, n):
4      if n == 0:
5          return 1
6
7      half = power(a, n // 2)
8
9      if n % 2 == 0:
10         return half * half
11     else:
12         return a * half * half
13
14
15  a = int(input("Enter base: "))
16  n = int(input("Enter exponent: "))
17
18  print("Result =", power(a, n))
```

Output

Output

Enter base: 2

Enter exponent: 5

Result = 32

=== Code Execution Successful ===

Time Complexity

- **Best Case: $O(\log n)$**
- **Average Case: $O(\log n)$**
- **Worst Case: $O(\log n)$**

Space Complexity

- **$O(\log n)$** (due to recursion)

Result

The **Exponentiation** algorithm was implemented successfully in Python using the **Divide and Conquer** technique. The program efficiently computes the power of a number and displays the correct result.